Chapter 3
System Preparation And Administration

First Steps

JBA recommend that you follow through Phases 1 to 6 and then, only when you have completed this successfully, continue with the Advanced System Manager Procedures and the Advanced Constructor Issues.

Warning: This section is for readers with a strong technical background.

These are the phases:

- Phase 1 Install Software To Server
- Phase 2 Configure The Server
- Phase 3 Install Software To PCs
- Phase 4 Start EWM Interactively From Application Manager
- Phase 5 Test Run EWM
- Phase 6 End EWM Interactively From Application Manager.

Phase 1 Install Software To Server

Note: Currently EWM supports the AS/400 server only. All JBA supported application servers will be supported in the future.

Step 1 System Manager

1. Install System 21 System Manager 3.5.2.
2. Install any 3.5.2 PTF versions of System Manager.

Note: You should now have an empty EWM file library which you will use later when you upload your Business Processes from the PC to the Server.

Step 2 Applications

1. If you plan to execute EWM then install the application software System 21 Applications, version 3.5.1 or a later version.
2. Note that you use EWM in two modes:

- Firstly Design Mode is where a Business Consultant uses Constructor to create or modify Work Management Business Processes, using a standalone PC connected to the Server via an ODBC connection. The consultant activates the Business Process and in so doing transfers it to the Server. The Business Process is now ready for execution and Constructor is no longer needed.
- Secondly Execution Mode is where System 21 users launch activities using Explorer. Note that Constructor is no longer needed.

Tip: Remember: You enable Activities but you activate a Business Process containing activities. An activity can be enabled without being part of a Business Process.

Step 3 ODBC

If you plan to use Constructor to Activate any Business Processes then the Server will need to be able to support ODBC connection from the PCs. So you need to set up the ODBC server software now.

Note: You set up the ODBC links from the PCs later.

Please do the following:

1. Use the command WRKSBS and check that the following subsystems have been started:
   - QSYSWRK
   - QSERVER

Note: If you wish you could use Machine Manager to enter any of the following commands automatically when the server starts.

2. If they have not been started then enter the following commands:
   - STRSBS SBSD(QSYSWRK)

   - STRSBS SBSD(QSERVER)

3. Enter this command:
   - STRHOSTSVR SERVER (*DATABASE)

4. Check that the following jobs are running in subsystem QSERVER.
   - QZDASRVSD
   - QPWFSERVSD
   - QSERVER

5. Check that either one or both of the following jobs are running in subsystem QSERVER:
   - QZDAINIT (where you are using Router ODBC)
   - QZDASOINIT (where you are using TCP/IP ODBC). If this is not running then enter STRTCP and then repeat STRHOSTSVR SERVER (*DATABASE).

6. The ODBC server software is now set up.

Phase 2 Configure The Server

Step 1 Authorisation Code

1. Ask JBA for a WF/V3 Authorisation Code for Work Management.
2. Use System Manager to enter the Authorisation Code.

Note: This alone will not enable EWM. It only says that the system may have environments in which EWM may be setup.

Step 2 EWM Environment Settings

1. Go into Application Manager and go to the Application Manager Menu.
2. Take Option 9 Enterprise Work Management.

Tip: As a shortcut you could enter STRIPGAM and 9 on a command line.

This displays a submenu with the following options:

- 1 Start EWM
- 2 End EWM
- 3 EWM Environment Settings
- 4 EWM Transaction Recovery
- 5 EWM Transaction Purge / Unlock

Step 3 Settings

1. Take Option 3 - EWM Environment Settings.
2. Enter the Environment you wish to enable. Note that in Phase 4 you will Start EWM in this environment.
3. For that Environment, enter the correct EWM Files library that Environment will use
4. Set the EWM Active flag.

Note: Only one Environment may reference a particular EWM Files library. A check is made to ensure multiple Environments do not attempt to share the same EWM files library. This prevents multiple EWM Engines accessing the same database and prevents cross-environment data corruption.

5. Take F20 to define a library list of Files libraries which will be used for Data Determination within that Environment. Manual Activities and Action Agents can use file fields to retrieve data. This library list will determine from which library the file will be accessed.

Note: This library list is mandatory for Active Environments.

6. Take F8 to update the setup.

Tip: To deactivate an Environment, simply set the Active flag to blank. It is not necessary to delete the Environment Settings record. In this manner, Environments may also be setup without actually activating them.

Phase 3 Install Software To PCs

Note: All PCs should be installed with Windows 95/NT.

Warning: Please consult JBA for the latest General Availability versions of Explorer and Constructor.

Step 1 Explorer

Install Explorer 3.5.2 on one PC for testing purposes. Refer to the Installation Guide within the Technical Information help file.

Tip: Install the remaining PCs when you have completed Phase 6.

Step 2 Constructor

Install Constructor 3.5.2 on one PC for testing purposes. Refer to the Installation Guide within the Technical Information help file.

Tip: After completing Phase 6 you then only need to install Constructor on those PCs which will be needed for designing or amending Business Processes.

Step 3 Server Connection

If you plan to execute EWM you need to:

1. Set up TCP/IP or Client Access SNA connection to the Server.
2. Set up an ODBC link to the server.

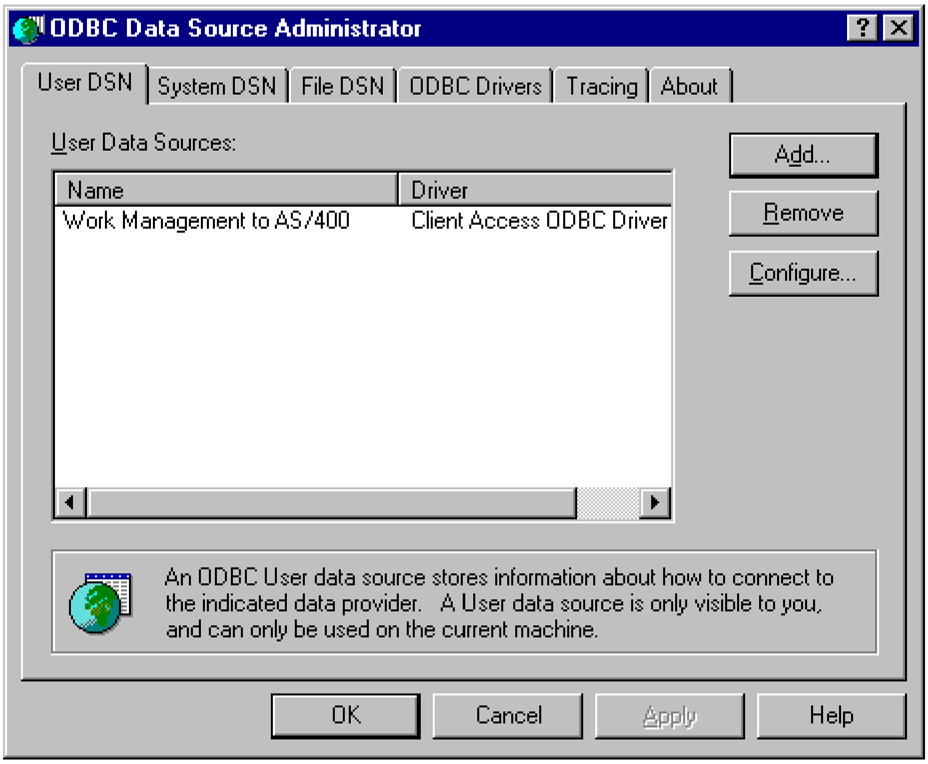
Set Up Connection To The Server.

You set up a TCP/IP or Client Access SNA connection to the Server in the normal way. There are no different or additional procedures needed here for EWM.

Setting Up ODBC Link

Note: JBA recommend that you consult your System Administrator because this can vary depending on your system. ODBC is usually found in Control Panel as shown below. Client Access sets this up as an icon in its own group rather than inserting it into Control Panel. You normally get ODBC with Windows NT, but not with Windows 95 by default. You may have it if you have installed Visual Basic or the Data Access options with MS Office, depending on which product options you selected for installation.

1. Launch ODBC and display the ODBC Data Source Administrator Window.



2. Note that there are three possible levels at which you can set these up and each use a corresponding panel on the Window:

   – User Level - Only available to the currently logged on user

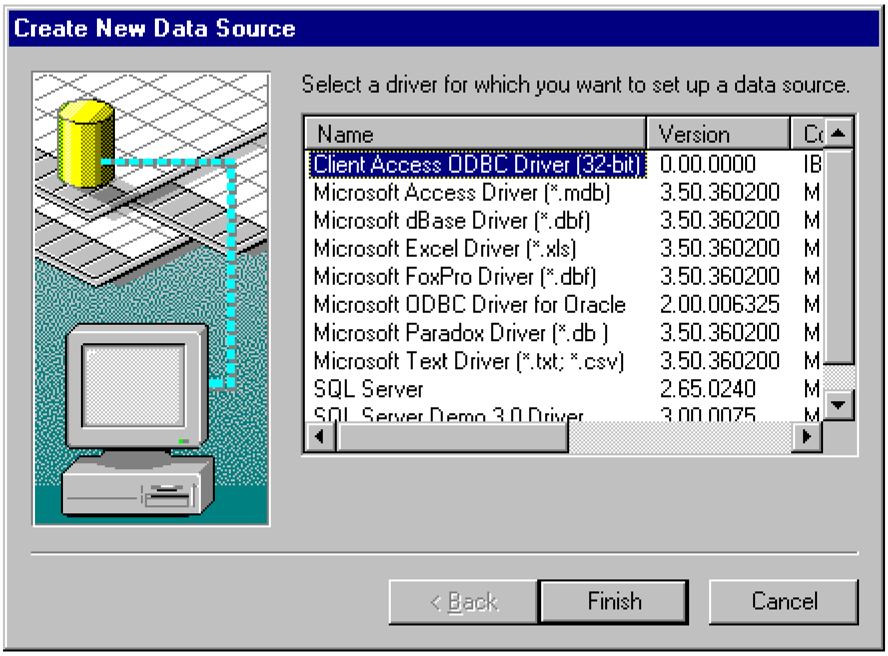
   – System Level - Available to all users logged on to the PC

   – File Level - Available to users with access to the required ODBC driver.

JBA recommend you to use User so you already have the correct panel selected here.

3. Select the Add button. This displays the Create New Data Source Window.

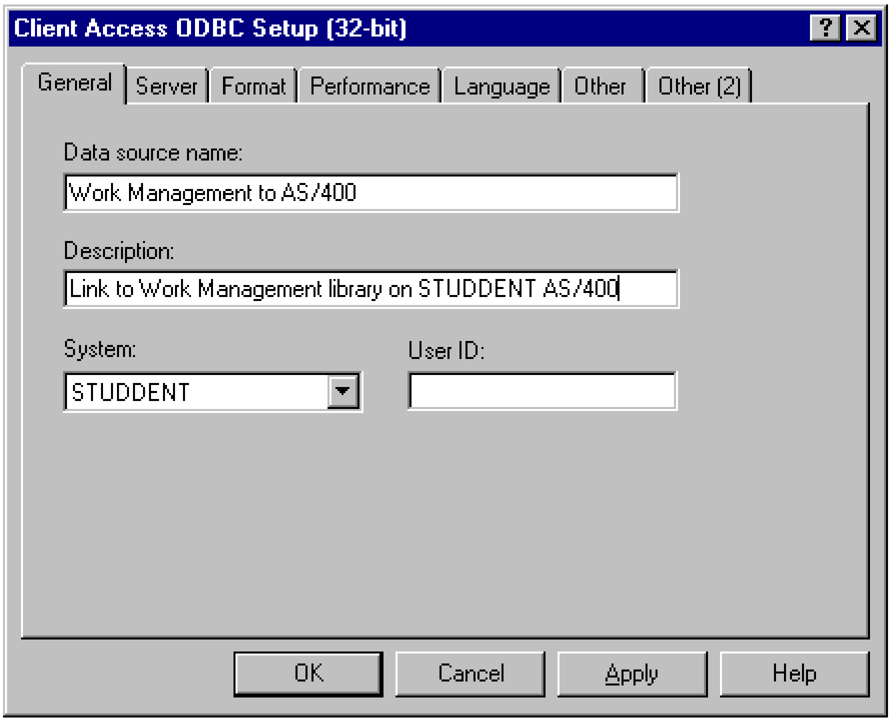
4. Select from the list the type of ODBC Driver required. This depends upon which drivers you have installed. In this example Client Access is assumed and this is the driver selected as follows.



5. Select the Finish button.
6. In this example the Client Access ODBC Setup (32-bit) Window is displayed. You complete the field details as follows. You use these four panels: General, Server, Performance and Other.

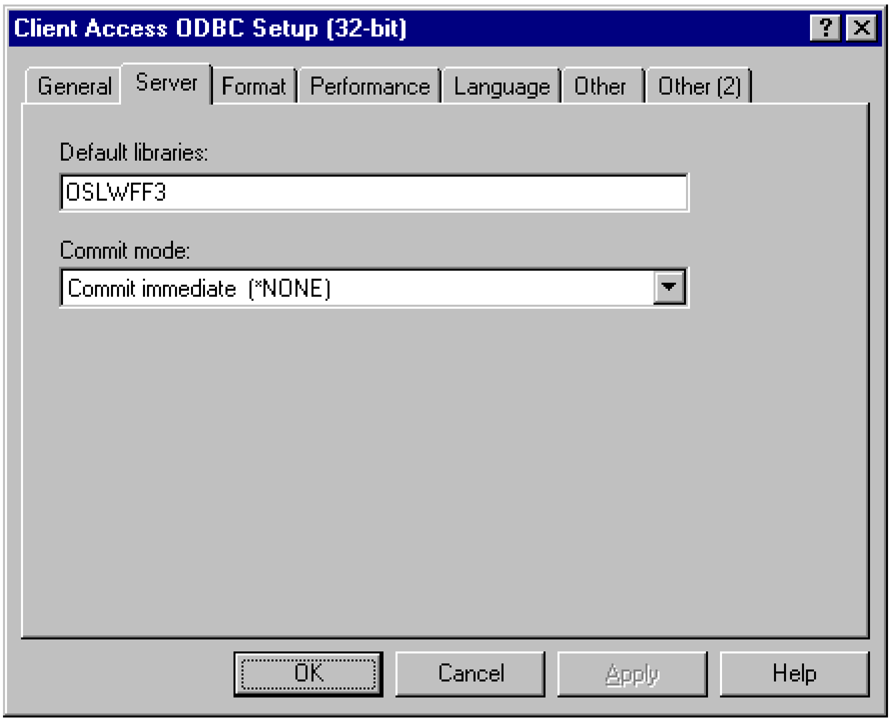
General

The Data Source Name entered is what will be shown in Constructor's Data Source Window which you see when you activate a Business Process.

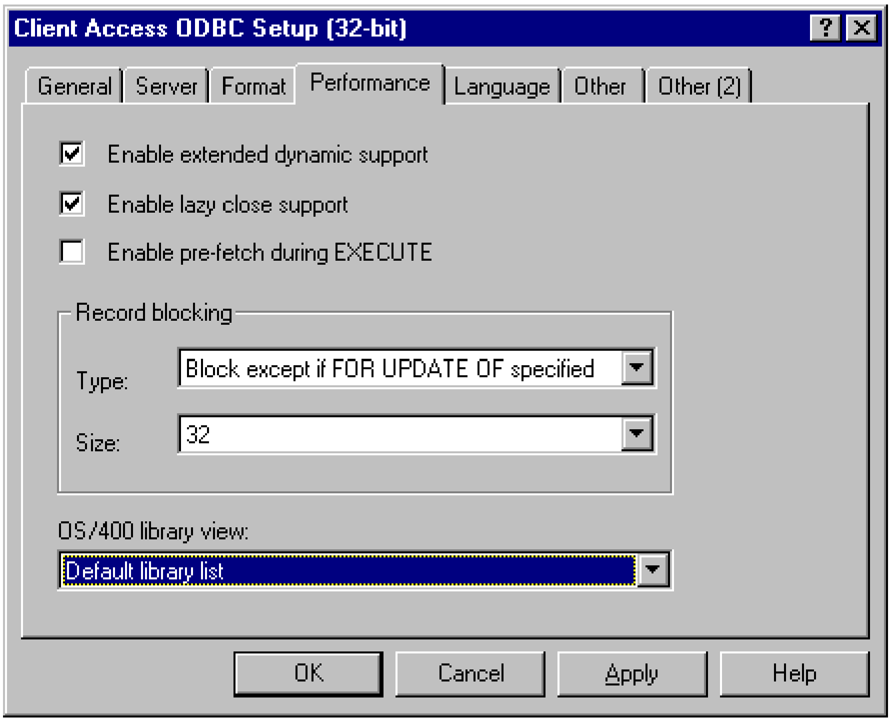


Server

Ensure that correct AS/400 Library name is specified. This is the library which contains the Work Management Database Files/Tables.

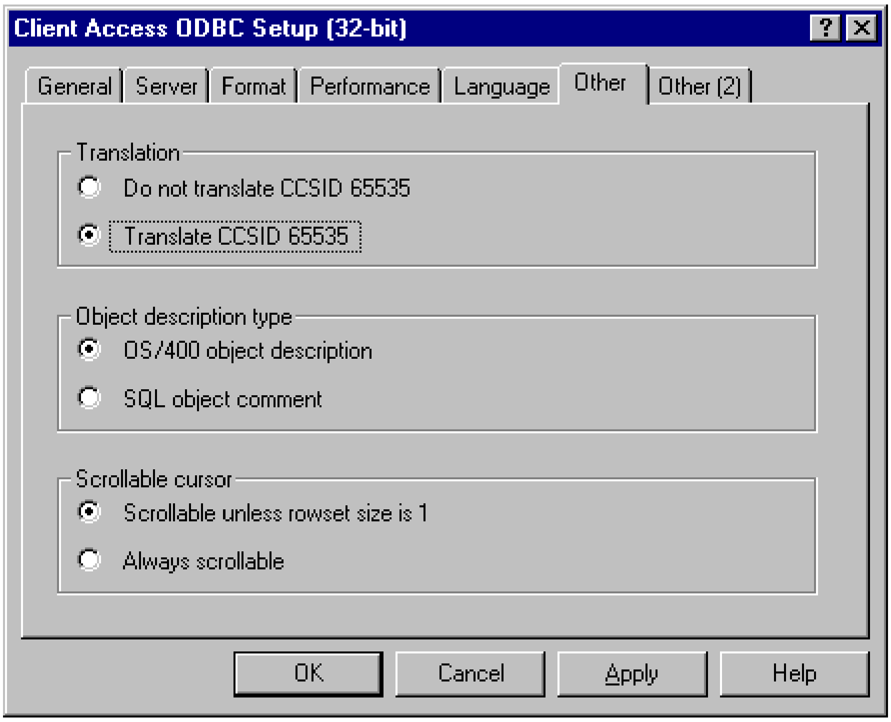


Performance



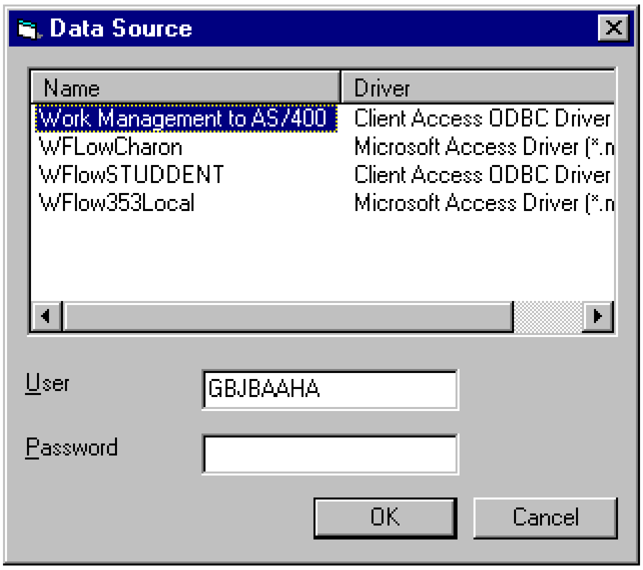
Other

Ensure that the Translation option is checked on. This ensures the correct ASCII to EBCDIC conversion.



This is the end of the ODBC procedure.

Note: When you activate a Business Process in Constructor you will be prompted to select the ODBC DSN as soon as a connection is required. Constructor displays the Data Source Window and the Data Source Name shown will be the one you have just entered during the course of this tutorial. This is the Data Source Window which you will see.



Phase 4 Start EWM Interactively From Application Manager

Step 1 Application Manager

Go into Application Manager and go to the Application Manager Menu.

Step 2 EWM Option

Take Option 9 Enterprise Work Management.

Tip: As a shortcut you could enter STRIPGAM and 9 on a command line.

This displays a submenu with the following options:

- 1 Start EWM
- 2 End EWM
- 3 EWM Environment Settings
- 4 EWM Transaction Recovery
- 5 EWM Transaction Purge / Unlock

Step 3 Start EWM

Take Option 1 Start EWM.

Warning: In order to start the EWM Engine the user must be fully authorised to run any application activity that the Engine might run automatically. Failure to be authorised will result in EWM Error records being written (to WFP97) and transaction records requiring Recovery.

Step 4 Settings

Enter the Environment you wish to start the EWM Engine in. Note that this must be the environment which you enabled in Phase 2. This will access the Environment Settings file, validate that the Environment is setup and active, before you are allowed to proceed.

A screen will prompt for any start up variation changes. The default setup will start the subsystem if required, then both EWM Engine and Scheduler. The options allow either jobs to be bypassed so that, if one was ended manually or abnormally, it could be restarted on its own. Another option allows the subsystem alone to be started.

The activity will then start the subsystem WFBACK3, if necessary, and then the EWM Engine and Scheduler, according to the confirmed options taken. Each job will have the EWM Files library defined on the Environment Setup added to the top of the System portion of the library list.

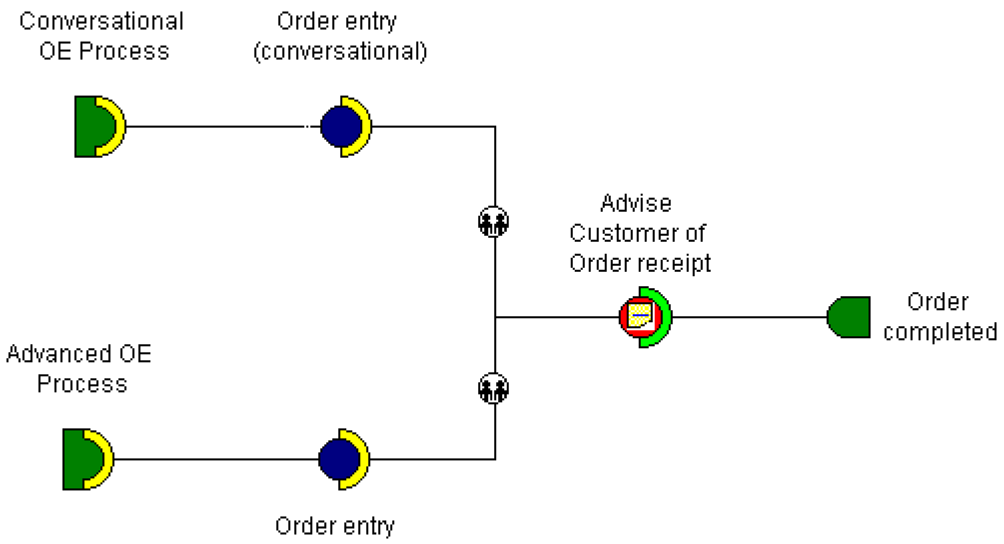
Phase 5 Test Run EWM

Step 1 Check Demonstration Data

Check that you have four processes which have been shipped as demonstration data: Ae0, Ae1, Ae2 and Ae3. This demonstration data is provided both as completed Constructor processes on the PC, ready for Activating, and also in the Work Management Engines main server data libraries as if they had just been activated.

Note: Ae1, Ae2 and Ae3 form the basis of the tutorials called Building AE Processes. However, you can ignore these for now and just concentrate on Ae0.

You are going to use Ae0 for testing that Work Management is functioning on the server. Ae0 is a very simple process which consists of Order Entry or Advanced Order Entry followed by a Manual Activity which is sent to the Sales Administrator. This is what Ae0 looks like on the Constructor Workspace.



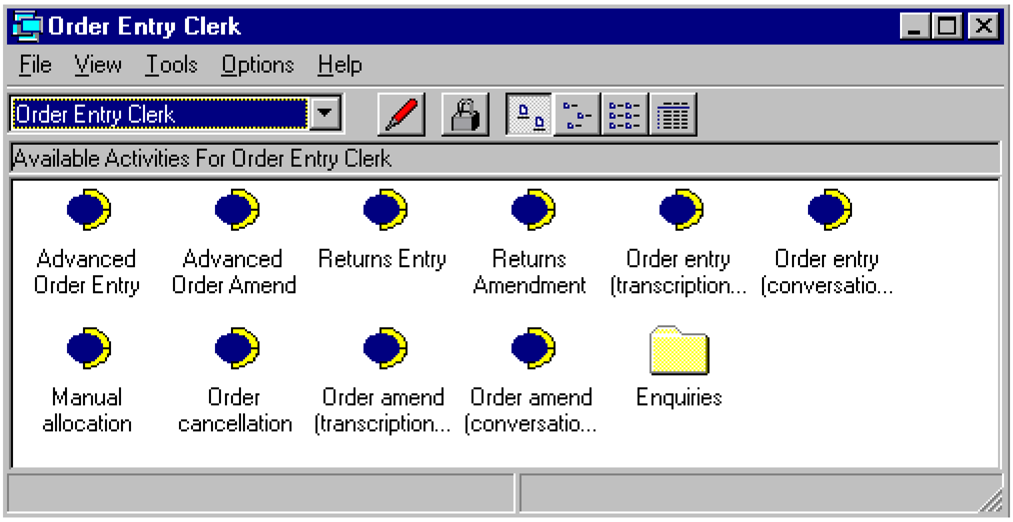
Step 2 Explorer

1. Go into Explorer. Note that Help is available from the Main Menu.

2. Check that you have the Action Tracker Toolbar Button. 

Note: If you cannot see this button then you must enable it using System Configuration. Select the Preferences option from the Options Menu and click the System Configuration button. Select the Settings panel and check the box called Enable Action Tracker Toolbar Button. Remember to select File\Save and then re-start Explorer.

3. Select the Order Entry Clerk role menu from the toolbar and display the activities as follows.



Note: You should have activities called Advanced Order Entry and Order Entry (Conversational). If you cannot see them please consult JBA for advice.

4. Double click either Advanced Order Entry or Order Entry (Conversational) to launch the Order Entry activity. Note that EWM does not become involved with the processing until the Order Number is created.

5. When you complete the Order the Work Management Engine will check to see which Business Processes contain Order Entry as the initial activity. Because you have four EWM enabled Business Processes: Ae0, Ae1, Ae2 and Ae3, all starting with Order Entry, EWM cannot determine which one to use. It should prompt you to decide which Business Process you want to use for this order. You should select Ae0.

Warning: If you are not prompted in this way then it means that one of the following statements are true and you cannot go any further until you correct the problem:

– You have not loaded the correct server PTFs

– EWM is not authorised (check that you have a WF/V3 Authorisation Code for Work Management)

– Some environment, library or user setting is wrong.

6. Note the Sales Order Number which you created.

7. Click the Action Tracker button to display the Action Tracker Window. 
This enables you to view the current status of any Business Object on your database.

8. Enter the Sales Order Number (remember the leading zeros), and you should see an Order Entry line (which has a green icon indicating it is completed) and then a Manual Activity line (with a red icon indicating it is waiting to run).

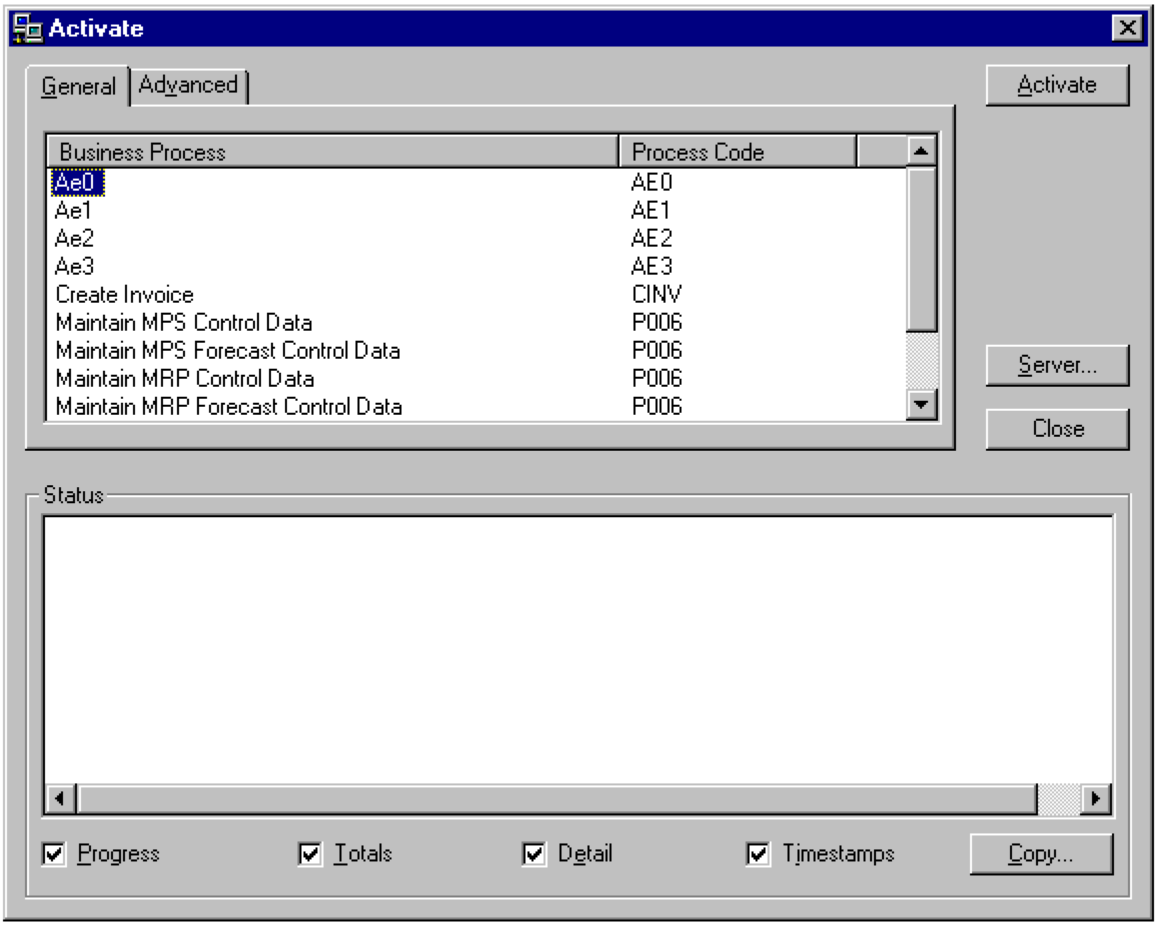
9. Switch to the Sales Administrator role in Explorer and turn on the Action List, using the button on the toolbar, and you should see the Manual Activity appear.

10. Double click on the Manual Activity in the action list and it should launch the activity. Note that a launched activity is greyed out.

11. If you then press the OK button on the Manual Activity and go back to the Action Tracker you should see that the Manual Activity line has a green icon and no other lines have appeared. If you have got this far you have successfully executed a Business Process.

Step 3 Constructor

1. Go into Constructor.
2. Select File\Open\Processes\Ae0.
3. Edit the Manual Activity called Advise Customer Of Order Receipt. Just for example change a word in the Message Panel.
4. Select the Activate option from the Work Management menu. This displays the Activate Window.

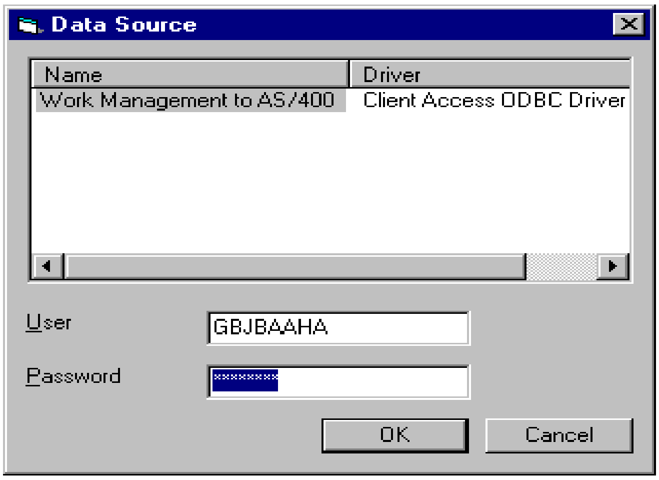


2. Scroll down the list, so that you can select Ae0, and then click Activate.

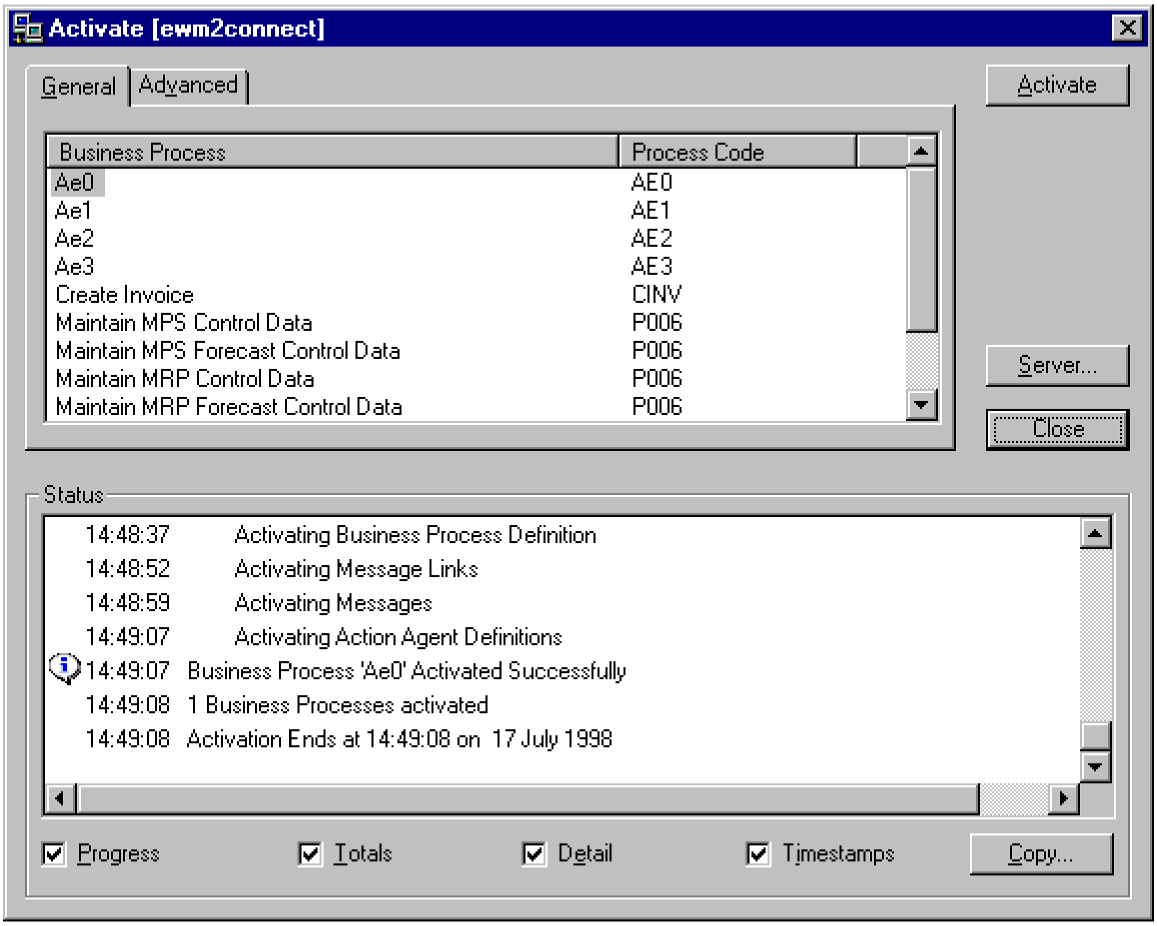
   All the components associated with a Business Process are held in the design-time database on your PC with the Business Process and when you activate all these components are saved on the Work Management files on the server with the Business Process.

   You can ignore the Advanced Panel.

3. You see the Validation messages in the Status box. If the validation fails with errors you will not be able to continue until you have corrected them. It the validation is successful it will say 'Validation ends'.
4. The SQL Data Sources Window is displayed for you to enter the database name. The one displayed here below is only an example. Check carefully which one you are to use.
5. You also enter a password and login.

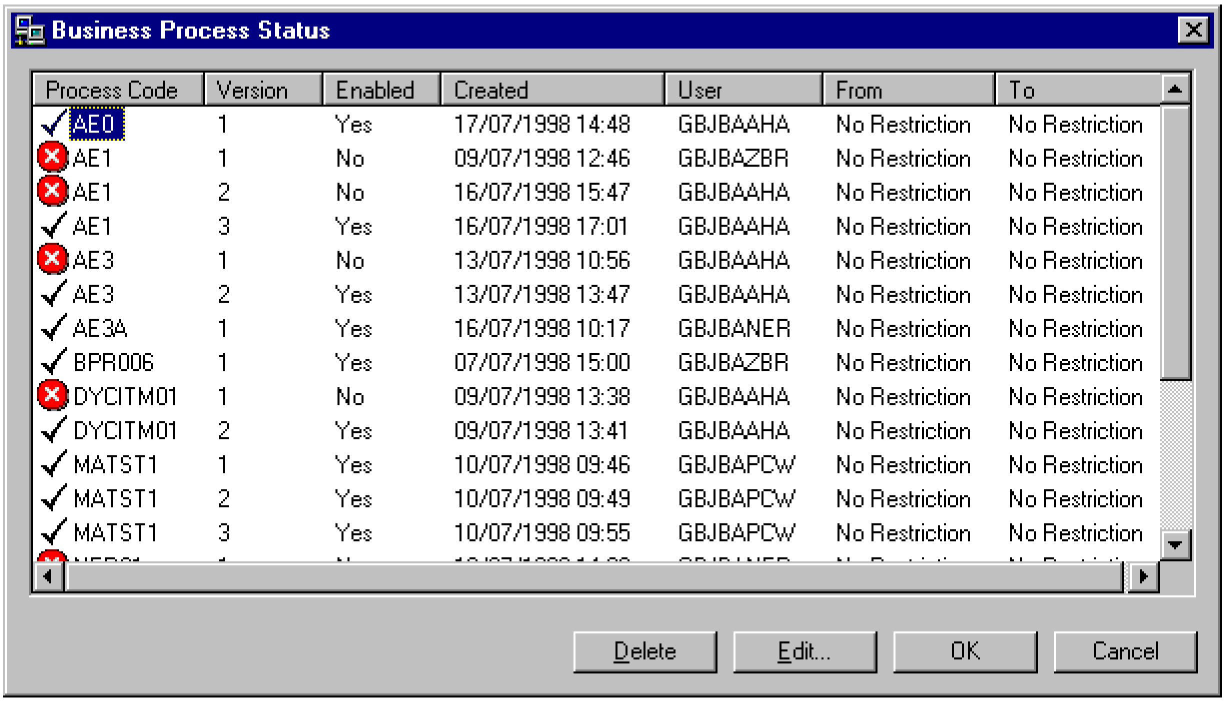


6. When the Business Process is activated you see the following window.



5. Ae0 is now loaded onto the server on WFP20.

   You can prove this by looking at the Business Process Status Window. To do this select Manage Activated Processes from the Work Management Menu. This window displays all the Business Processes which are activated to the server. You should see Ae0 at the top of the list.



Step 4 Explorer Again

1. Go into Explorer.
2. Create another order.
3. If you see your new modified Manual Activity message appearing you know that you can successfully create and/or modify processes in Constructor and then Activate them to the server.

Phase 6 End EWM Interactively From Application Manager

Step 1 Application Manager

Go into Application Manager and go to the Application Manager Menu.

Step 2 EWM Option

Take Option 9 Enterprise Work Management.

Tip: As a shortcut you could enter STRIPGAM and 9 on a command line.

This displays a submenu with the following options:

- 1 Start EWM
- 2 End EWM
- 3 EWM Environment Settings
- 4 EWM Transaction Recovery
- 5 EWM Transaction Purge / Unlock

Step 3 End EWM

Take Option 2 End EWM.

Step 4 Settings

Enter the Environment in which you wish to end the EWM Engine. This will verify that the EWM Engine or Scheduler are currently active in the chosen Environment.

A screen will prompt for any end variation changes. The default setup will end both EWM Engine and Scheduler, and then the subsystem if required. The options allow either jobs to be bypassed, so that one may be ended manually.

Note: Another option allows the subsystem alone to be ended.

The activity will carry out the confirmed selection.

Advanced System Manager Procedures

Take note of the following issues only when you have successfully completed the First Steps towards preparing and administering EWM.

Starting And Ending EWM From Machine Manager

Note: This is for the AS/400 only.

EWM may be started and ended by including commands to control EWM in the Auto Day Start (ADS) and/or Auto Day End (ADE) functions of Machine Manager.

Note: EWM must be setup in System Manager prior to starting EWM via Machine Manager.

The STRWM and ENDWM commands will carry out the same process as described above, but will bypass all interactive display processing. A valid environment must be passed as a parameter. The default variation will be used, and any errors will be trapped and displayed in the job log.

Note: The User that submits the STRWM job must be fully authorised to run any application activity that the Engine might run automatically. That is the STRWM job is as detailed in Machine Manager, or set by a job description assigned to the STRWM activity. Failure to be authorised will result in EWM Error records being written (to WFP97) and transaction records requiring Recovery.

Start And End Issues That May Arise

Environment fails validation

Check the Environment Settings for that Environment. There must be a record, it must have the Active flag set to '1', and a valid library containing EWM files as its library.

Start EWM Engine or Scheduler: Couldn't start - activity already running

A check is made against the EWM Record Lock (WFP99) file in the EWM files library defined against the chosen Environment. If a record exists for the Engine or Scheduler, the respective activity will not be started. Run the following command (AS/400 only) to verify what jobs are running in the subsystem:

WRKSBSJOB WFBACK3

Place a '5' against any job, and check the library list to determine which files library (and therefore with a check against the Environment Settings file (WFP02) - which Environment) is in use.

If the lock is genuine, then you have attempted to start a job that is already running. If there are no active jobs for the chosen Environment, but a lock is on the WFP99 file, then the associated activity ended abnormally. In order to free this, please refer to the Unlock utility.

End EWM Engine or Scheduler: Couldn't end - activity not running

A check is made against the EWM Record Lock (WFP99) file in the EWM files library defined against the chosen Environment. If a record does not exist for the Engine or Scheduler, the respective activity will not be ended.

Run the following command (AS/400 only) to verify what jobs are running in the subsystem:

WRKSBSJOB WFBACK3

Place a '5' against any job, and check the library list to determine which files library (and therefore with a check against the Environment Settings file (WFP02) - which Environment) is in use.

If there are no active jobs for the chosen Environment, then there is nothing to end - you have attempted to end a job that is not running. If there are active jobs for the chosen environment, confirm that there are no appropriate WFP99 lock records in the relevant EWM files library. This should never occur in normal use. The recommended way to clear this is to end any EWM jobs that may be ended normally, and then end the subsystem as follows:

ENDSBS WFBACK3 ENDOPT(*IMMED)

Once completed, run the Unlock utility to clear any spurious records and then the next restart the EWM Engine and Scheduler should function correctly.

Start or End does not activate

Check that there are no jobs on the WFBACK3 job queue. The WFBACK3 subsystem is configured as shipped to allow two jobs to run (one EWM Engine, one EWM Scheduler). If multiple Environments are to be EWM activated, then the maximum number of jobs allowed in the subsystem should be adjusted accordingly. This is a System Administrator task and should be referred accordingly.

EWM Engine or EWM Scheduler appears to have ended abnormally

Confirm that just the one job has failed. If there is still a record on the relevant lock file (WFP99) for the activity, and the subsystem has no active task, the job may have ended abnormally. It may have been ended manually too (option 4 from WRKSBSJOB or WRKACTJOB).

If the subsystem is no longer running, but lock records remain, then an ENDSBS WFBACK3 ENDOPT(*IMMED), or even PWRDWNSYS with a low wait time may have been issued.

The EWM Engine and Scheduler will detect a shut down of the subsystem or system if a wait period of over 300 seconds (5 minutes) is specified. It will then end normally. Otherwise it is ended by outside control, and the locks will remain.

To remove locks, refer to the Unlock Utility.

EWM Transaction Recovery

System Manager offers a recovery routine for EWM transactions that either failed EWM validation, or EWM automatic batch submission. To view the transactions that require recovery:

1. Take option 9/4.
2. Enter the Environment you wish to start the EWM Engine in. This will access the Environment Settings file, validate that the Environment is setup and active, before you are allowed to proceed.
3. The detail screen shows:
   - All records that were cancelled whilst on a job queue
   - All records that appear to be in active state, yet do not have an active job associated with it.

The latter may have ended abnormally, or have been ended manually, or have been ended whilst in progress by an ENDSBS or PWRDWNSYS command.

4. To recover a single transaction, select it and take F8. This will back out the current EWM transaction process and reset it ready to run that transaction process again. If the transaction is a batch activity, it will resubmit the activity to process.

   Note: This will only occur for jobs cancelled on the job queue. Unreconciled active jobs need manual checking. The status will be ready to rerun the current process, but it should be run manually after investigating why it failed the previous time.

5. It is possible to process all cancelled job types in one go, by taking F16.

EWM Transaction Purge And Unlock Utility

This utility allows a clearance of EWM transaction data, and/or a clearance of the lock file.

Warning: This will not touch the data activated by JBA Constructor, referred to as master data.

1. Take option 9/5.

Warning: Care must be taken when using this utility - it is intended for unlocking previous completed jobs (a cleanup exercise), or for clearing test data. Clearing live data may impact on EWM transactions processed afterwards. Currently, there is no completed EWM transactions archive - a transaction purge will lose EWM transaction history.

Tip: It may be possible to clear all EWM transactions through, and then to archive the Files library to media. To save space, the transaction purge may be run, and then a fresh fileset will be used by future EWM transactions.

Tip: Always backup the EWM files library prior to a transaction purge.

2. Enter the Environment in which you wish to start the EWM Engine. This will access the Environment Settings file, check that the Environment is setup and active, before you are allowed to proceed.

Note: All EWM processing in the chosen Environment must be ended prior to running this option.

3. Select the options to process. Selecting the Transaction Purge will clear the transaction files. Selecting the Unlock will clear the Lock file (WFP99). The master data files updated by JBA Constructor are not affected by this utility.

Advanced Constructor Issues

Take note of the following issues only when you have successfully completed the First Steps towards preparing and administering EWM.

EWM Engine Performance And Sizing

The Work Management Engine has been designed to minimise the overhead placed on the Server. At this point in time we have no quantitative data on the effect of running Work Management processes.

The impact on performance is mainly influenced by two factors:

- The volume of documents under Work Management control. For example, how many Sales Orders, Pick Notes and Invoices are processed by an Order Entry process.
- The complexity of the process definition. In particular the substitution of variables will have the most significant impact as each variable substitution requires at least two SQL statements to be executed. For example you use variable substitution when using text in a manual activity.

As a rough guide we would not expect the impact of Work Management to be significant if the following factors are true:

  - With low volumes of document processing, for example up to 30 orders per hour
  - With an average process complexity, for example Ae3
  - On a Server that is providing adequate interactive performance.

Activity Control

When you run a Work Management enabled activity some additional operations occur on the Server as each document is created. For example the activity could be Order Entry, in which case the document created would be an order. This is what happens:

- Application Manager checks if Work Management is authorised.
- Application Manager checks for the activity in other processes. Work Management needs to know whether the activity that created the document, is the first or Initial Activity in any of the other processes which have been activated on the Server and if so then Work Management needs to know how many other processes are involved:
  - If this activity is not the Initial Activity in any Business Process then the document is not created under Work Management control.
  - If this activity is the Initial Activity in only one Business Process then the document is created under control of the given process.
  - If this activity is the Initial Activity in more than one Business Process then the user is presented with a window requesting them to select which Business Process they want to use to control the order.

Note: If you reduce the number of enabled Business Processes for any given initial activity, down to one then you will not get this window prompting the user to select a process. Processes can be enabled and disabled using the Manage Activated Processes option on the Work Management Menu in Constructor.

Warning: You can remove this window but please use extreme caution. Do not do this without advise from your JBA consultant. All you need to do is to look in WFP20 and remove from here all the Business Process files, except the one you want.

Tip: This is important: Users do not execute a Business Process, they execute activities in the same way whether Work Management exists or not. It is the Server that recognises that the selected activity initiates a Business Process. The only user involvement is where the system cannot determine which Business Process to execute where more than one has been defined in relation to the executed activity.

Key Elements In Constructor

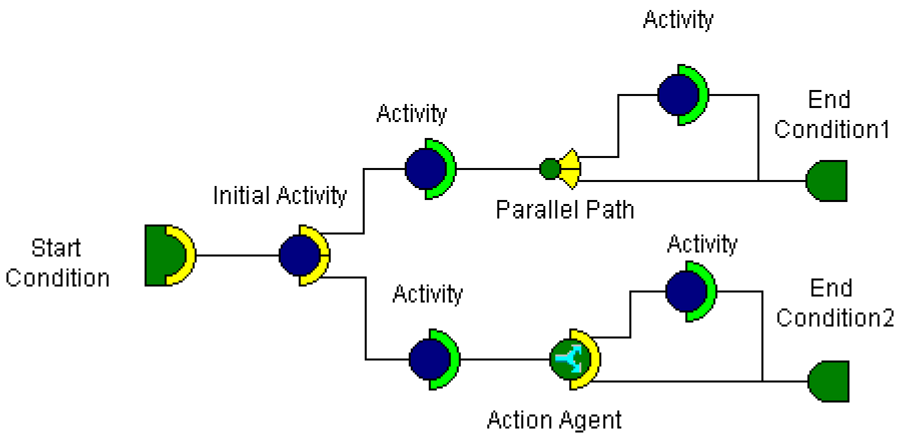
There are key elements in Constructor models which are needed by Work Management:

- Start Condition
- End Condition
- Initial Activity - This decides the Business Process.
- Activity:
  - This has a Document Type defined which is associated with the output and the next input
  - Completion codes control the next step
  - Execution Modes can be defined for each activity. The Work Management Engine needs to know whether this is user invoked or automatic.
- Parallel Paths
- Action Agents

Note that a Business Process may have multiple Initial Activities. If an activity is connected to a Start Condition symbol in Constructor , then it is an Initial Activity.

The following picture shows the main components of a Business Process and how they link together on the Workspace in Constructor.

Note: For detailed information see the Tutorial help section called Building AE Processes.



Initial Activities

An activity can be included in the designs for more than one Business Process. The EWM Engine does not recognise that the activity is enabled for Work Management until the order is created, and at this point the Business Process is not known and may vary according to the order created.

When the Explorer user executes a System 21 Activity, for example, Order Entry, the Work Management Engine makes the following checks to make sure that the correct Business Process is being used:

<u>Check 1</u>

When the order is created the Engine checks that the activity is enabled for Work Management. If the activity is not enabled the processing continues without Work Management control of any sort. If the activity is enabled then Work Management takes control of the processing and then makes Check 2.

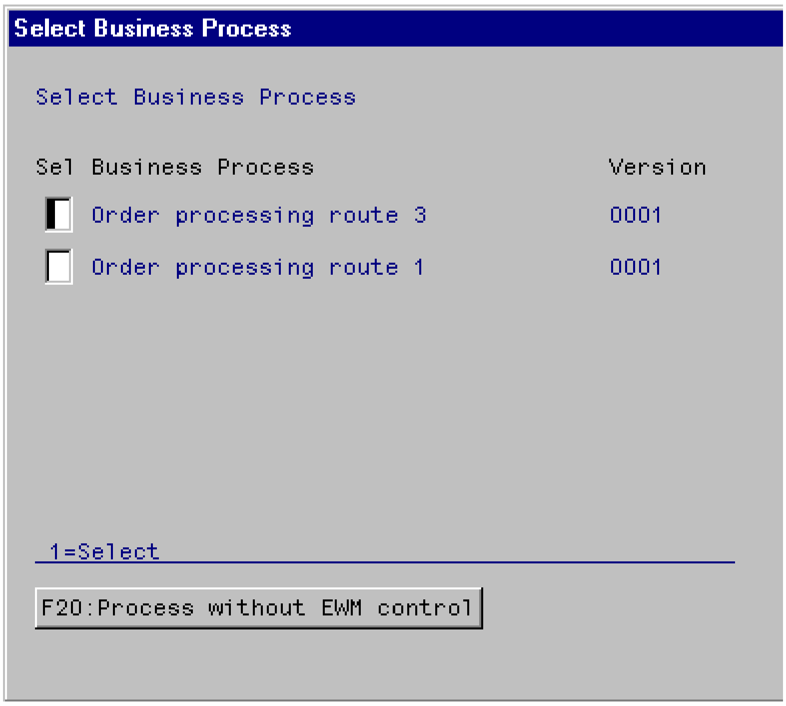
<u>Check 2</u>

An activity can be included in the designs for more than one Business Process, for example, Order Entry so, for this reason, the Engine checks all the Business Processes to find the ones for which this activity in question is the Initial Activity.

It then displays a list of all these Business Processes and the Explorer user selects the appropriate process.

When the Business Process has been selected the Work Management Engine knows how to process the activities within this Business Process. The processing now follows Constructor's design for the Business Process.

In the following example the Explorer user created an order using the Order Entry activity. The Engine recognised that this activity was the initial activity for two Business Processes and displayed the two Business Process names and version numbers. The user could choose between the two Business Processes or Select F20 to work outside of the Work Management control altogether.


Select Business Process
Select Business Process
Sel Business Process Version
Order processing route 3 0001
Order processing route 1 0001
1=Select
F20:Process without EWM control

Version Control

Identification

The Business Process Code uniquely identifies the Business Process to the Work Management Engine, whereas it is the name of the Business Process which uniquely identifies the Business Process to Constructor and to the design-time database on the PC.

When you work with a Business Process on Constructor's Workspace you can change the Code using the Properties option on the Workspace pop up menu. You can then save the Business Process, with the new Code, to the design-time database on the PC.

Activation

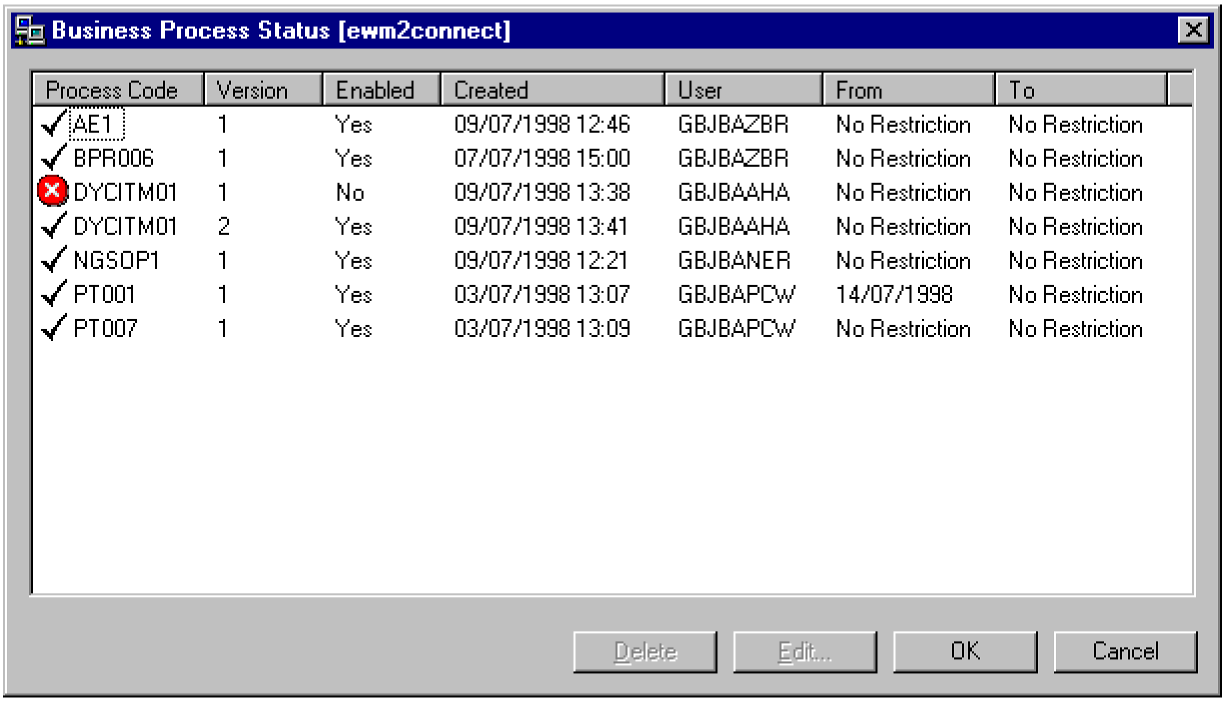
However, if you Activate the Business Process the Work Management Engine checks the Business Process Code, not the Business Process name, and saves the Business Process to the Work Management tables on the server, where the Business Process Code is now the key to the Business Process. These details are saved on WFP20 where the version number is the PVER20 field.

If the Business Process Code is new to the Work Management tables then a new file is created with a version number 001. However, if that Business Process Code already exists the Work Management Engine creates a new file for the Business Process but increments the version number. In this way all Business Processes are version controlled on the server, as identified by the Code, whereas only the latest copy of the Business Process is held on the PC and is identified by its name.

Management

You can use the Business Process Status Window to manage the activated business processes on the server. This is what you can do:

- You can view the Business Process details, including the version number and effectivity dates. There is a row of details for each version of the Business Process.
- You can change the status of the Business Process to disabled or back again to enabled. If it is disabled this means that although the Business Process exists on the server, it is not considered by the Work Management Engine for possible initial activities, although it will complete any processes which it has already started under its control.
- You can define effectivity dates, between which the Business Process will be enabled or disabled.
- You can delete a Business Process version.



Process Code	Version	Enabled	Created	User	From	To
AE1	1	Yes	09/07/1998 12:46	GBJBAZBR	No Restriction	No Restriction
BPR006	1	Yes	07/07/1998 15:00	GBJBAZBR	No Restriction	No Restriction
DYCITM01	1	No	09/07/1998 13:38	GBJBAAHA	No Restriction	No Restriction
DYCITM01	2	Yes	09/07/1998 13:41	GBJBAAHA	No Restriction	No Restriction
NGSOP1	1	Yes	09/07/1998 12:21	GBJBANER	No Restriction	No Restriction
PT001	1	Yes	03/07/1998 13:07	GBJBAPCW	14/07/1998	No Restriction
PT007	1	Yes	03/07/1998 13:09	GBJBAPCW	No Restriction	No Restriction

This is a typical display showing all the Business Processes which have been activated to the server.

<u>Explorer</u>

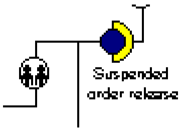
The only time an Explorer user is aware of the Business Process, under which control the activities are running, is when they execute an activity which is the initial activity for more than one Business Process. The Work Management Engine displays a list of all these Business Processes and the Explorer user selects the appropriate process. The version number is displayed with the name of the Business Process.

Defining Execution Modes

There are Execution Modes which are already defined for you to use within Constructor. You can also define others yourself.

You select the Execution Mode on the Select Schedule Window in Constructor. The symbol you see on this window is then placed in front of the Activity, for which you are adding scheduling details, on the Constructor Workspace.

For example, here the activity called Suspended Order Release has a User Controlled Schedule.

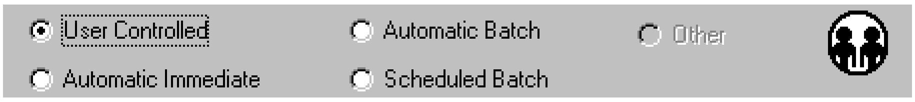


The shipped Execution Modes are as follows:

blank

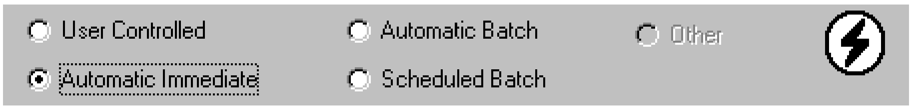
Business Object document not previously under control of the EWM Engine. The activity was started by the User. This is common for Initial Activities.

10 User Controlled.



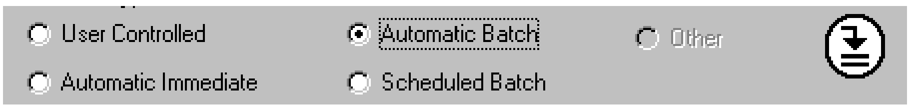
The User controls when this activity will be run. This usually means that the activity will be on an Action List until the User selects it.

20 Automatic Immediate



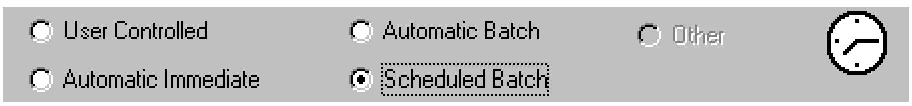
The EWM Engine will process this job in batch, and put it onto a job queue normally setup for immediate processing. To this end, the job queue associated with this mode by default is QINTER in QGPL.

30 Automatic Batch



The EWM Engine will process this job in batch, and send it to a common batch queue for the purpose. The default as shipped is QBATCH in QGPL.

40 Scheduled Batch



All Scheduled activities are passed by the EWM Engine and handled by the EWM Scheduler.

Other

The Other option is available only if you have defined your own execution modes. Please refer to the online help available with Constructor. You can choose your own symbol for the Workspace.

Note: You can define an Email schedule, a User Schedule and a Role Schedule. These require Data Fields to be set up first. Please refer to the Adding Data Fields section of this help guide.

Activities Supported

Only the activities listed, in the appendix called Enabled Activities List, are currently enabled for Work Management in the standard product.

It is possible to use non-Work Management enabled activities as supporting activities to a manual activity which is part of a Work Management process. In this case key data, for example Order Number, can be passed into the activity to save re-keying it.

Please refer to the tutorials for rebuilding Ae2 and Ae3 in the help section called Building AE Processes.

Multiple Activities In One Process

A current restriction of the Work Management enabled activities, listed in the appendix called Enabled Activities List, is that any of those activities may appear only once in a Work Management Enabled process. Where a Work Management Enabled process calls other nested child processes these are all considered to be part of the main process and so likewise may not duplicate an activity which is in the parent or another child process.

Note: This restriction does not apply to Action Agents, Parallel Paths or Manual Activities or any other activity which has been enabled following the guidelines in this guide.

Manual Activity Restriction

When entering manual activity text, the number of characters that can be entered, including for example Data Field definitions, Button definitions, through Constructor is unlimited. However on the server, the amount of text that can be processed in a manual activity message is limited to 960 characters (including data fields, etc).

Warning: If this limit is surpassed, the end of the message will be cut off.

Anyone entering new manual activity messages should be aware of and test for this restriction after creating their messages. In future versions of the product it is planned that this restriction will be eliminated.

Defining Data Fields Using SQL

There is a limitation when defining Data Fields in Constructor using SQL.

Warning: Please ensure that the SELECT statement only tries to select one field per Data Field. Any other fields will not be returned by the Dynamic SQL in WF401.

For example, Customer Postcode is made up of two five-character fields on SLP05 (PCD105 and PCD205). Interactively, SQL will return both fields when the following statement is run :

```
CustPostCode   SELECT PCD105, PCD205 FROM SLP05
```

If this statement is defined in Constructor, only PCD105 will be returned (even if the return field is 10 characters long).

For this reason, in order to represent Postcode fully, two data fields are required, defined as follows :

```
CustPostCode1     SELECT PCD105 FROM SLP05.......
CustPostCode2     SELECT PCD205 FROM SLP05.......
```

To include the full Postcode in a Manual Activity message, two data fields must be included in the definition of the message. This is due to limitations to Dynamic SQL in RPG.

Escalation And Delegation

Note: Please refer to the Escalation And Delegation Window in Constructor.

Activities that require manual intervention or messages are sent to the appropriate users desktop through the Action List feature in Explorer.

The Action List shows a list of all the activities that the user should carry out. This list is sorted by priority. The actual list is a combination the following:

- All Action List items for the current role that Explorer is showing, as per the Role combo box.
- Items for the current user, changed by the Padlock button on the toolbar.

The title of the Action List window shows the Role, and User if one has been selected, that the Action List is for.

If items are ignored and not processed from the Action List, the Business Process may be stalled. It is therefore important that it is possible to move items, either by changing their priority, so they appear higher up the Action List, or by re-assigning them to some other Role or User.

This process is called Escalation and Delegation.

The difference between Escalation and Delegation is that the action of changing the priority/recipient is a function of time. For example, Escalation is where the item has been sitting on an Action List for *n* hours, where *n* is a numeric value. Whereas Delegation occurs immediately and is typically used where the intended recipient is absent through holidays or sickness.

Configuration of the Escalation and Delegation rules is done through Constructor. This is then updated on the Server during the Activation process. Escalation and Delegation rules are executed periodically on the Server by checking all Action List items and matching them to the Escalation and Delegation rules.

Rules may be defined at various levels. There are:

  – User Rules
  – Role Rules
  – Process Rules
  – Activity Rules
  – General Rules.

For example, all these rules could be set up as follows:

- A user who is away on holiday might have a delegation rule set up to re-direct all Action List items to another user. (User Rule).
- Another rule may be set up to escalate any activity within a given process after two hours. (Process Rule).
- Another rule may be set up to escalate all activities in all processes after eight hours. (General Rule).

Activation Performance

You activate a Business Process by copying it from its definition in Constructor to the Work Management Engine's working file set. This is also referred to as Upload.

Tip: Refer to the last task in the tutorial for Building Ae3, in the Building AE Processes help topic.

During this process a considerable amount of data is transferred. This data transfer is very SQL intensive and can take a considerable time. Activation times of several minutes for one Business Process are not unusual.

During Activation of processes, all dependent data for that process is also transferred to the server: this includes Schedule Rules, Data Fields, Execution Modes and Document Types. The only additional data items that must be Activated to the server are the Escalation And Delegation Rules.

Defining Codes

When defining Business Processes, Role Codes and Activity Codes it is quite possible, even desirable to use long descriptive names. For example, My Order Entry Process, Sales Administration, Confirmation Of Despatch. However, to minimise traffic on the Work Management Engine, Business Processes, Role Codes and Activity Codes are known to the Server by much shorter names.

Warning: If you provide Business Processes, Role Codes and Activity Codes with identical codes then it will not function as you intended.

This is an easy mistake to make. For example, you might for instance decide to create a new Order Entry process by modifying and re-naming an existing one. However, if you omit to change the Process Code, by right clicking on the Start Node and changing the Process code field then the existing one will be overwritten by the new one when it is activated.

Process Code

This is the name on the screen but it is also known as Business Process Code. Change this by right clicking on the Start Node of a process in Constructor.

Warning: If you change this and it is not unique then the existing one will be overwritten by the new one when it is activated.

Role code

Change this by using the Role Codes option from the Options Menu in Explorer. If this is not unique, all those roles sharing the same Role Code will share the same Action List. Note: If you change a Role Code then you must re-activate all processes that use it as they pass the current Role Code to the Server when activated.

Activity Code

Change this by editing the Properties of an Activity within Constructor using the Activity Code field on the Work Mgt panel.

When enabling an activity for Work Management you must assign a unique 10 character Activity Code.

Warning: Failure to do so could cause the Business Process to fail unpredictably. If the code is not unique the Work Management Engine could pass control to the wrong user.

The Activity Code is a 10 character alpha string and must be configured both within the source code of the activity and within the representation of that activity in Constructor.

You change this code by editing the Properties of an activity in Constructor using the Activity Code field on the Work Mgt panel.

However, this code is also hard coded into the RPG program. You must not change the Activity Code in Constructor without changing the same code in the RPG. Also, all processes that use the activity will have to be re-activated.

This is how the Activity Code is made up:

Char	**Purpose**
1-3	Originator. This will be the JBA Customer Code for field modified programs or 'JBA' for standard product.
4	Action Style. This is used to signify different methods of carrying out the same action verb. If only one style exists then use 'S' for Standard. Examples: A Advanced, C Conversational, T Telesales.
5-7	Action Verb. Examples: ENT Enter, GNR Generate, UPD Update, CRT Create, PST Post, AUD Audit, RCV Receive, PRT Print, DEL Delete.
8	System Type. Example: G Generic, D Drinks, S Style
9-10	Document Type. Example: SO Sales Order, PO Purchase Order, IN Invoice, PN Pick Note.

For example, the Telesales Order Entry activity, within Drinks, produced and shipped as a JBA standard Work Management enabled activity, would have an Activity Code of:

JBATENTDSO.

Where: JBA means JBA Activity, T meansTelesales, ENT means Enter, D means Drinks, SO means Sales Order.

Note: For historic reasons the Work Management enabled activities listed, in the appendix called Enabled Activities List, do not follow this convention.

Button Field Overrides

In Manual Activities, buttons may be defined which allow the user to run another activity in support of the Manual Activity message (e.g. an enquiry).

If the activity to be launched is a system 21 activity where key data must be entered (e.g. Customer), then this can be automatically entered by the system as the activity is launched by defining field overrides. These are set on the overrides tab on the button definition dialogue when setting up the message for a Manual Activity within Constructor.

When defining field overrides, the field number is specified together with the value to be inserted into that field. The value can be either static text, or the value of a data field.

Warning: If the value is a data field, then the field itself must appear as part of the Main message text, that is a field appearing in red in the main body of the message.

Multi Language Support

Constructor currently uses only UK English.

Warning: You can enter Activity Titles and Manual Activity messages in a foreign language but font support may not be correct. This will be resolved in due course.